

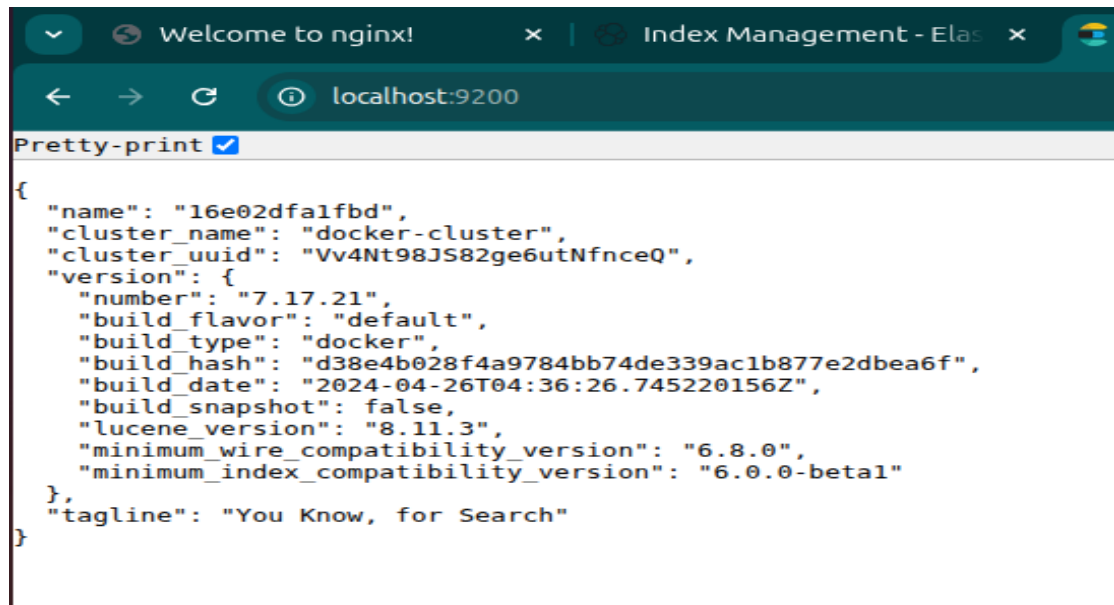
Docker Assignment

Name : A Avinash Reddy (TSV700)

Elasticsearch – Filebeat – Kibana Stack Monitoring Nginx Logs

This documentation explains the implementation of an ELK stack integrated with Filebeat to monitor Nginx logs using Docker Compose. The project demonstrates centralized logging where Nginx access and error logs are collected, shipped to Elasticsearch, and visualized using Kibana. Docker containers are used to isolate each component, ensuring modular deployment. The stack consists of Elasticsearch for storage, Kibana for visualization, Filebeat for log shipping, and Nginx as the log source. All services are connected through a custom Docker network. This setup reflects a production-style monitoring architecture.

1. Docker Compose Configuration

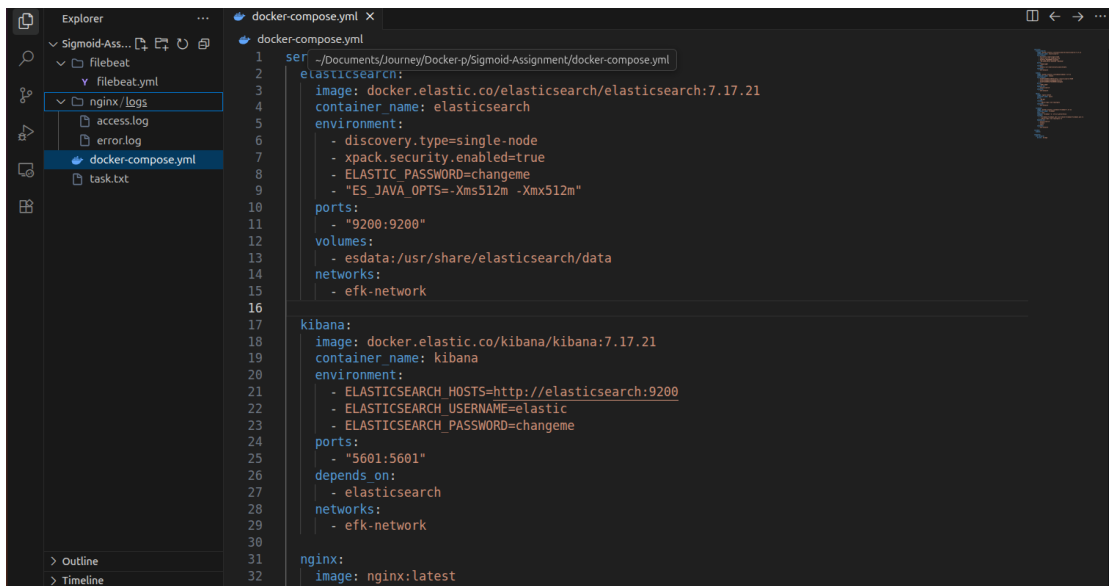


```
{
  "name": "16e02dfaf1fbd",
  "cluster_name": "docker-cluster",
  "cluster_uuid": "Vv4Nt98JS82ge6utNfncEQ",
  "version": {
    "number": "7.17.21",
    "build_flavor": "default",
    "build_type": "docker",
    "build_hash": "d38e4b028f4a9784bb74de339ac1b877e2dbea6f",
    "build_date": "2024-04-26T04:36:26.745220156Z",
    "build_snapshot": false,
    "lucene_version": "8.11.3",
    "minimum_wire_compatibility_version": "6.8.0",
    "minimum_index_compatibility_version": "6.0.0-beta1"
  },
  "tagline": "You Know, for Search"
}
```

The docker-compose.yml file defines all services required for the ELK stack. Elasticsearch is configured in single-node mode with security

enabled. Kibana is connected to Elasticsearch using environment variables. Nginx runs as the web server generating access and error logs. Filebeat is configured to monitor Nginx log files and forward them to Elasticsearch. Port mappings expose services such as 9200 for Elasticsearch and 5601 for Kibana. A shared Docker network allows internal communication between containers. Volumes are used to persist Elasticsearch data and share Nginx logs with Filebeat.

2. Running Containers Verification



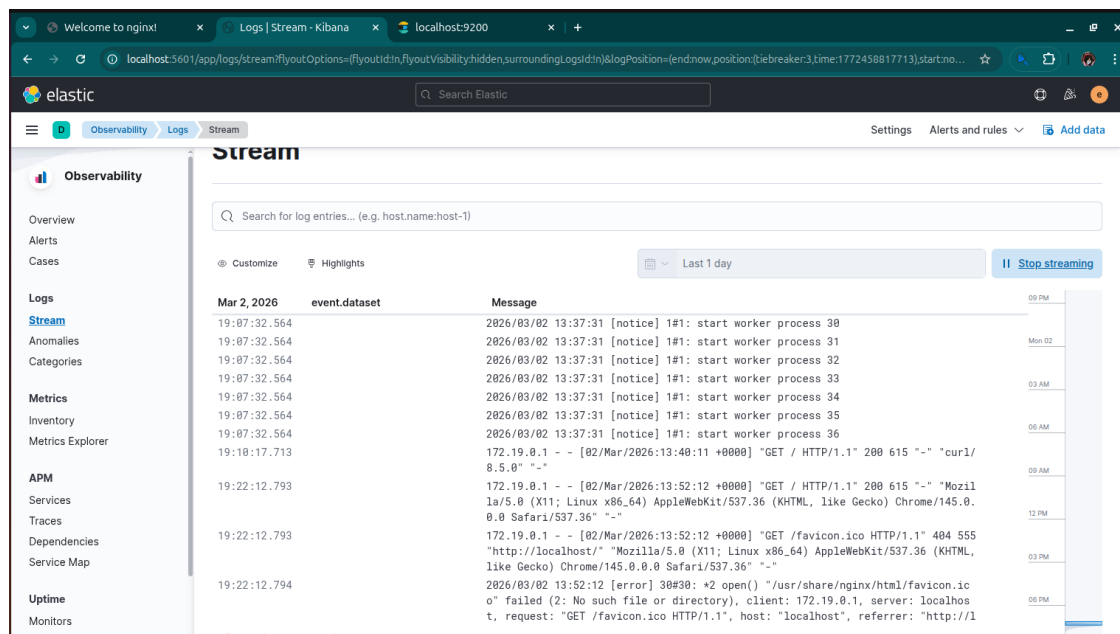
The docker ps command confirms that all containers are running successfully. The output shows Elasticsearch, Kibana, Filebeat, and Nginx containers in active state. Port mappings verify external accessibility of services. Elasticsearch is exposed on port 9200, and Kibana on port 5601. Nginx is accessible on port 80 for generating logs. Container IDs and image versions confirm correct Elastic Stack version 7.17.21. This step validates that Docker Compose deployment executed properly. It ensures the monitoring stack is operational before log inspection.

3. Elasticsearch Health and Index Verification

```
signoid@sv-lap-0108:~$ cd /home/signoid/Documents/Journey/Docker-p/Sigmoid-Assignment
signoid@sv-lap-0108:~/Documents/Journey/Docker-p/Sigmoid-Assignment$ ls
docker-compose.yml  filebeat  nginx  task.txt
signoid@sv-lap-0108:~/Documents/Journey/Docker-p/Sigmoid-Assignment$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
b2d3cc223295   docker.elastic.co/beats/filebeat:7.17.21   "/usr/bin/tini -- /u..." 13 minutes ago Up 13 minutes
cfb3c420c9cf   docker.elastic.co/kibana/kibana:7.17.21    "/bin/tini -- /usr/L..." 13 minutes ago Up 13 minutes   0.0.0.0:5601->5601/tc
p, [::]:5601->5601/tcp
22f3a247eace   nginx:latest                             "/docker-entrypoint..." 13 minutes ago Up 13 minutes   0.0.0.0:80->80/tcp, [
::]:80->80/tcp
16e02dfa1fbd   docker.elastic.co/elasticsearch/elasticsearch:7.17.21   "/bin/tini -- /usr/L..." 13 minutes ago Up 13 minutes   0.0.0.0:9200->9200/tc
p, [::]:9200->9200/tcp, 9300/tcp
signoid@sv-lap-0108:~/Documents/Journey/Docker-p/Sigmoid-Assignment$ curl -u elastic:changeme "http://localhost:9200/_cat/indices?v"
health status index      uuid      pri rep docs.count docs.deleted store.size pri.store.size
green open   .geoip_databases  Tvs20T3CQpInZPNY7_4j0A 1 0 41 0 39.6mb 39.6mb
green open   .security-7      KblwW-TuSP0Zz6g0UqExw 1 0 53 0 269.4kb 269.4kb
green open   .apm-custom-link jgQY5Ij55AKMGjP6-wjAA 1 0 0 0 227b 227b
yellow open   filebeat-7.17.21-2026.03.02 LxVkoCCQ3aRfW70FK-5Q 1 1 18 0 29.8kb 29.8kb
green open   .apm-agent-configuration xUtCdPfpRvOUY52ytAR2rQ 1 0 0 0 227b 227b
green open   .kibana_7.17.21_001 zQJL5bz7QZyW3SHs4qPxnQ 1 0 320 55 2.4mb 2.4mb
green open   .kibana_task_manager_7.17.21_001 49Xw0t7LSxKlYrFB-JLsw 1 0 18 2517 374.8kb 374.8kb
signoid@sv-lap-0108:~/Documents/Journey/Docker-p/Sigmoid-Assignment$
```

Accessing <http://localhost:9200> confirms that Elasticsearch is running. The JSON output displays cluster name, version, and build details. This verifies successful initialization of the Elasticsearch node. Security is enabled using the elastic user credentials. The `_cat/indices` API command lists all created indices. The `filebeat-*` index confirms that logs are being ingested. Index status shows green or yellow health depending on shard allocation. This step proves successful log storage within Elasticsearch.

4. Kibana Log Visualization



Kibana provides visualization of logs stored in Elasticsearch. The Logs Stream interface displays real-time Nginx access and error logs. Each entry includes timestamp, dataset, and message details. Log streaming confirms Filebeat is correctly forwarding log data. The interface allows filtering and searching through log entries. Observability features help analyze application and server behavior. Kibana dashboards can be created using filebeat indices. This completes the centralized logging pipeline from Nginx to visualization.